

Movers Transport System

Project Plan

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1. PROJECT OVERVIEW

Movers is a transport company that moves agricultural produce and farm supplies between farms and retail outlets. Daily work spans farmer registration, trip scheduling, fleet and staff management, cost and payment tracking, and driver discipline. Much of this is still handled manually or across disconnected records, which slows scheduling, makes costs hard to reconcile, and leaves disciplinary history scattered.

We propose a relational database system backed by a web application that centralizes these operations in one place. The database is the source of truth for who we serve, what moves on each trip, what it costs, and how drivers are held accountable. A role-based front end lets staff and customers interact with that data without exposing raw tables. The goal is fewer errors, faster trip assignment, clearer financial records, and consistent enforcement of company policy.

2. WHAT THE SYSTEM DOES

The system supports the full operational lifecycle of Movers transport:

- Register large-scale farmers individually and small-scale farmers through groups (minimum five members per group).
- Manage vehicles, drivers, and loaders, including availability and assignment to trips.
- Create and track trips with origin, destination, distance, cargo type, load weight, and assigned resources.
- Calculate transport cost, apply tax, and record customer payments per trip.
- Log fuel use and vehicle servicing tied to trips and fleet.
- Record driver offences, issue warning letters, apply suspensions, and support termination when policy thresholds are met.
- Provide role-appropriate views so each user sees only what they need for their job.

3. WHO IT SERVES

Primary users are internal Movers staff and external farmer customers. Each role has a defined responsibility:

#	User Role	Primary Responsibility
1	System Admin	Full Access
2	Ops Manager	Vehicle and Driver Management
3	Accountant	Financial transactions

4	HR Manager	Driver discipline & staff
5	Driver	View own trips & records
6	Farmer/Customer	Request trips, view payments

Farmers are the paying customers. Large-scale farmers use the system as individuals; small-scale farmers participate through registered farmer groups. Retail outlets are served indirectly as trip destinations. Internal roles depend on accurate, shared data, so scheduling, finance, and HR stay aligned.

4. MAJOR ENTITIES INVOLVED

The database is organized around five areas of the business. The entities below are the main tables; relationships follow standard relational design (e.g., farmers link to groups through membership, trips link to vehicles, drivers, loaders, and payments).

Farmer Management

- Farmer —> stores information about individual farmers
- FarmerGroup —> manages groups of small-scale farmers
- GroupMembership —> links farmers to their groups

Fleet and Staff

- Vehicle —> stores transportation vehicle information
- Driver —> stores driver records and employment information
- Loader —> stores loader information for transportation activities

Trip Operations

- Trip —> stores transportation trip information
- TripLoader —> links loaders to transportation trips
- Payment —> stores transportation payment information

Vehicle Maintenance

- FuelRecord —> tracks fuel usage and cost
- ServiceRecord —> stores vehicle maintenance information

Driver Discipline

- Offence —> records driver violations
- WarningLetter —> tracks formal warnings
- Suspension —> records suspension actions

Together, these entities support the main operations of the Movers transport system.

5. BUSINESS RULES

The database and application will enforce the following rules:

- Large-scale farmers register individually; small-scale farmers must belong to a group with at least five members.
- Every trip requires exactly one driver and one vehicle.
- Every trip requires at least one loader (recorded via TripLoader).
- Transport cost is calculated from base rate, distance, and load weight; tax is 20% of transport cost; total cost is transport cost plus tax.
- Three surcharges on a driver result in a warning letter.
- An offence after a warning results in suspension.
- Two suspensions within six months result in termination.
- A vehicle may only be assigned to a trip when it is available; a driver may only be assigned when active and not suspended.

6. FRONT-END PLAN

The front end will be a web application designed to help users interact with the system through role-based dashboards. Users will log in and access tools and information based on their responsibilities.

Planned screens by role

Farmer / Customer

- Registration —> individual farmer signup or create/join a farmer group.
- Request transport —> submit origin, destination, cargo type, and load weight.
- My trips —> status and history of requested trips.
- Payments —> view the amount owed and the payment status per trip.

Operations Manager

- Fleet —> list vehicles with status (available, assigned, in transit, maintenance, etc.).
- Staff —> drivers and loaders; check who is active and available.
- Assign trip —> create a trip, pick a vehicle, a driver, and at least one loader; enter route and cargo details.
- Trip list —> search and open trip details.

Accountant

- Trip costs —> transport cost, tax, and total per trip.
- Payments —> record payment method, date, and amount against a trip.
- Fuel and service —> review FuelRecord and ServiceRecord entries linked to trips and vehicles.

HR Manager

- Offences —> log offence type, date, and surcharge for a driver.
- Discipline history —> warnings, suspensions, and counts toward termination policy.
- Driver status —> see who is suspended or at risk under policy rules.

Driver

- My trips —> assigned routes, dates, and cargo summary.
- My discipline —> own offences, warnings, and suspensions (read-only).

System Admin

- User and role management —> assign roles to system accounts.
- Overview —> high-level access to modules other roles use, for support and configuration.

Primary user flow

A typical trip flows as follows: a farmer or group requests transport from the customer portal; an operations manager assigns an available vehicle, an active unsuspended driver, and at least one loader, and the system calculates cost and tax; the accountant records payment when received; fuel and service costs are logged as needed; if a driver commits an offence, HR records it and the system supports tracking toward warning, suspension, or termination per business rules.

Users will log in and be directed to role-based dashboards that provide access to the tools needed for their responsibilities. The interface will prioritize simple navigation, clear forms, and organized views of system data.

7. CONCLUSION

The Movers Transport System gives the company a single, reliable place to manage farmers, trips, fleet, payments, and driver discipline. The system captures the main operations and business rules used by Movers while making daily activities easier to manage through a centralized platform. We are ready to implement this design and demonstrate it in the project review meeting.